Robotous Force Torque Sensors

RFT40-SA01
Mini
Lightweight

RFT44-SB01
Mini
Dust Free

RFT60-HA01
Thin
High Resolution

RFT64-SB01
Thin
Dust Free

RFT64-6A01
High Resolution
High Reliability

RFT80-6A01
High Resolution
Easy Mounting

RFT80-6A02
High Resolution
High Reliability

RFT80-6A03
High Torque
High Reliability

The Technology Enabling Physical AI

Powered by innovative capacitance-based technology, the RFT Series delivers advanced multi-axis force/torque sensing and real-time performance - enabling humanoids, robots, and AI systems to interact with the physical world with greater intelligence, precision, and safety.

RFT Series – Key Technological Highlights

Capacitive Sensing
Revolutionizing force measurement with unparalleled precision and efficiency

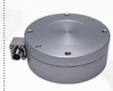
High Resolution
Detects even the finest force and torque changes

Temperature Compensation
Ensures reliable operation in harsh thermal environments

Overload Protection
Safeguards both the sensor and robotic systems, ensuring long-term reliability and safety

All-in-One with Digital Output
Simplifies integration and enables real-time data processing

Specification Table

Item		Unit	RFT40-SA01	RFT44-SB01	RFT60-HA01	RFT64-SB01	RFT64-6A01	RFT80-6A01	RFT80-6A02	RFT80-6A03
Dimension	Diameter	mm	40	44	60	64	64	80	80	80
	Height	mm	18.5	20	18.5	20	30	22	30	33
Data Rate	(maximum)	Hz	200	200	1,000	1,000	1,000	1,000	1,000	1,000
Load Capacity	Fx, Fy	N	100	100	150	150	200	400	400	1,000
	Fz	N	150	150	200	200	200	400	400	1,000
	Tx, Ty, Tz	Nm	2.5	2.5	4	4	10	20	20	50
Resolution	Fx, Fy	mN	200	200	100	150	80	100	80	200
	Fz	mN	200	200	150	200	80	100	80	200
	Tx, Ty, Tz	mNm	8	8	5	5	4	5	4	10
Overload Capacity	Fx, Fy	%	150	150	150	150	300	150	300	300
	Fz+ (Fz-)	%	150 (300)	150 (300)	150 (300)	150 (300)	300 (300)	200 (400)	300 (300)	300(300)
	Tx, Ty, Tz	%	150	150	150	150	300	200	300	300
Hysteresis	Fx, Fy	%FS	2	2.5	2.5	3	1	2.5	1	2
	Fz	%FS	0.5	1	1	2	1	1	1	2
	Tx, Ty, Tz	%FS	1	3	1	2	1	1	1	2
Cross Talk	Fx, Fy	%FS	3	3	3	3	2	3	2	1.5
	Fz	%FS	3	3	3	3	2	3	2	1.5
	Tx, Ty, Tz	%FS	3	3	3	3	2	3	2	1.5
Weight(w/o cable)		g	60	70	120	140	175	226	294	322
Power Requirement		W, VDC	5V(24V for EtherNET, EtherCAT)							
Output Interface		Choices	CAN | RS232 | RS422 | USB | EtherCAT | RS485 | EtherNET							
Product Image										
Feature	Hollow Core ○ / Dust Seal ●		-	●	○	●	-	○	-	●
	Temperature Compensation		-	-	-	-	●	-	●	●
	Embedded EtherCAT Option		-	-	-	-	●	-	●	●